Experiment -2

NAME: MADHUMITHA P

ROLL NO:2407011297

Chess Game UI – Instruction Page Analysis

NSTRUCTION PAGE: Chunking Analysis of the Instruction Page Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The Memory Recall Task instruction page effectively utilizes chunking in the following ways

1. Clear and Sequential Numbering

- The instructions are broken down into six steps, making them easy to follow.
- Each step presents one key action in a structured manner, reducing cognitive overload.

Example Steps:

1. Set up the chessboard with pieces in their starting positions.
2. White moves first.
3. Each player alternates turns.
4. Capture opponent's king to win.
5. You will have a fixed time per move (optional).
6. Accuracy in moves is more important than speed.

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2. Logical Grouping of Information

- **Observation Phase (Steps 1-2):** Users learn about the board setup and who moves first.
- **Strategy Phase (Step 3):** Encourages players to plan moves carefully.
- **Gameplay Phase (Steps 4-5):** Explains how the game progresses and introduces time limits.
- **Key Performance Metric (Step 6):** Highlights that accuracy matters more than speed.

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3. Visual Hierarchy and Design Elements

- Bold, large title ("CHESS GAME") → Grabs attention and clearly states the task.
- Bullet points and spacing → Reduce clutter, enhancing readability.

- Highlighted “START GAME” button → Signals the next step, keeping navigation intuitive.
- Chess-themed visuals (board, pieces) → Add a professional yet engaging touch.

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4. Time Constraint Reinforcement

- Step 5 explicitly states, “You will have a fixed time per move.”
- This reinforces expectations while subtly urging users to focus.

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5. Simplicity and Clarity

- Instructions use short, direct sentences, avoiding unnecessary complexity.
- The active voice makes it more engaging and action-oriented.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

- **Clear Numbering:** Instructions broken into steps (setup, rules, start).
- **Logical Grouping:** Explains objective, piece movement basics, and start conditions.

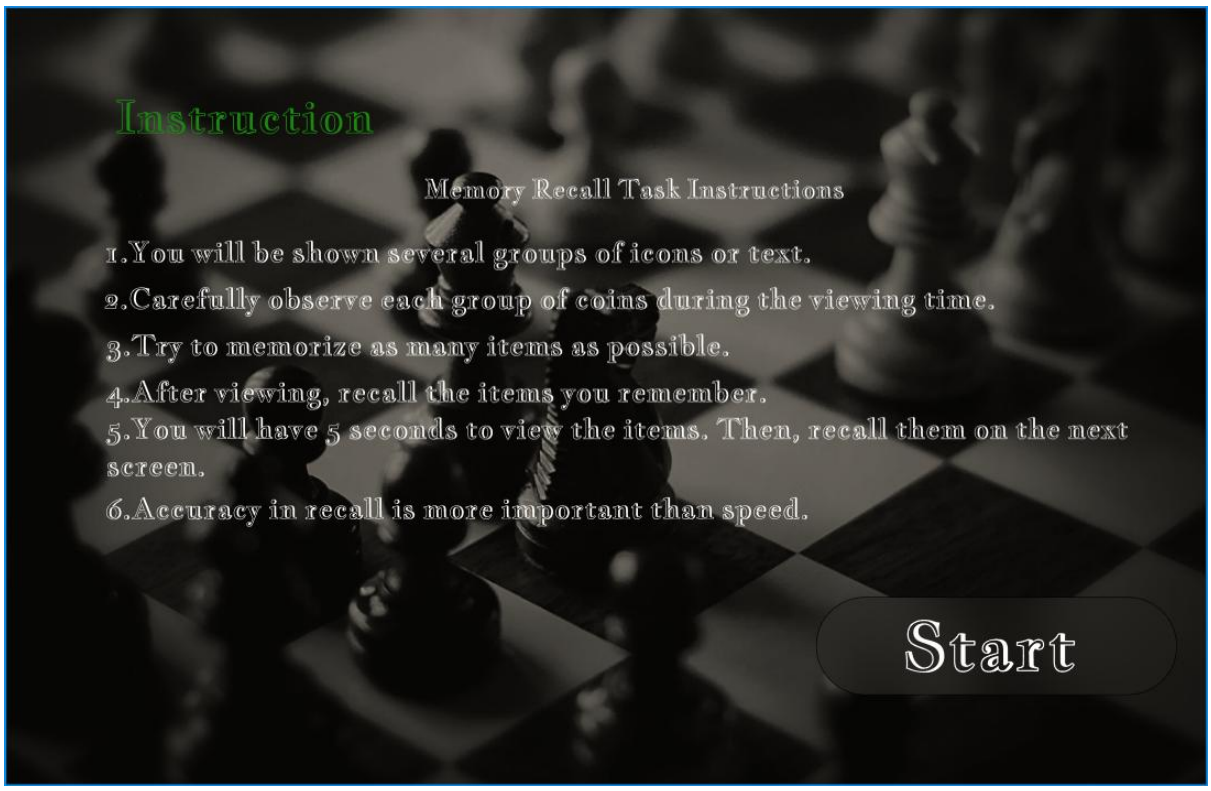
- **Visual Hierarchy:** Bold title “CHESS GAME”; highlighted “START GAME” button.
- **Time Constraint Reinforcement (optional):** If timer is used, mention it here.
- **Simplicity & Clarity:** Short, direct sentences, active voice.

Instructions Example:

1. Set up the chessboard with pieces in their starting positions.
2. White moves first.
3. Each player alternates turns.
4. Capture the opponent’s king to win.
5. Accuracy in moves is more important than speed.

START GAME

FRAME 2: GAMEPLAY PHASE



Analysis of Gameplay Screen

- **Purpose:** Main interaction phase where players move pieces.
- **Key UI Elements:**

- Chessboard grid (8x8).
- Interactive pieces (drag-and-drop or click-to-move).
- Timer (optional).
- Move history panel (records previous moves).

- **Cognitive & UX Benefits:**
  - Clear grid reduces confusion.
  - Move history aids recall and strategy.
  - Timer adds urgency for competitive mode.

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FRAME 3: CHUNKING PHASE

1. Purpose of the Screen

- **This is the visual memory encoding phase, where users observe and mentally group chess pieces before recalling them.**
- **The term "Chunking Phase" suggests that the game applies the chunking principle, helping users remember items by categorizing them into meaningful groups.**

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2. Key Elements and UI Components

- **Countdown Timer (Top Left – Green Circle: “00.05”)**
    - **Indicates users have 5 seconds to observe and memorize the chess pieces.**
    - **The green color and bold format create urgency and focus.**
- **Progress Bar (Top Center – Grey Horizontal Bar)**
    - **Visually represents time remaining.**
    - **Reinforces urgency and helps users track progress.**
- **Grid of Chess Pieces (Two Rows)**
    - **Includes a mix of white, black, and brown pieces such as pawns, rooks, knights, bishops, and queens.**
    - **Visually distinct pieces encourage chunking strategies:**
        - **By type (e.g., all pawns, all rooks)**
        - **By color (white, black, brown)**
        - **By position or repetition (e.g., multiple white pawns or black rooks)**
- **Background Theme**
    - **Blurred chessboard background adds thematic depth without distracting from the task.**

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3. How the Chunking Phase Works

1. **Users scan the grid and look for patterns or related pieces.**
2. **The countdown timer limits observation time, forcing quick memory strategies.**
3. **Once time is up, users transition to the Recall Phase, where they must identify previously seen pieces.**

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4. Cognitive and UX Benefits

- **Enhances short-term memory by allowing users to recall groups of pieces instead of individual ones.**
- **Reduces cognitive overload by helping users organize data efficiently.**
- **Improves pattern recognition, making recall easier and more accurate.**
- **Visually engaging layout with chess themes makes the task professional and intuitive.**

FRAME 4: SELECTION PHASE

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1. Purpose of the Screen

- This is the **memory retrieval stage**, where users select chess pieces they remember from the previous timed display.
- The goal is to test short-term memory and recognition accuracy.

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2. Key Elements and UI Components

Title & Instructions

- "CHESS GAME" (Bold Header) – Reinforces the purpose of the task.
- "SELECT THE PIECES YOU REMEMBER:" – Clear instruction guiding the user.

Grid of Chess Piece Choices

- Eight chess pieces are displayed in a grid format.
- Includes both previously shown pieces and **distractors** (new pieces added to test memory).
- Each piece has a **green dot** below it, indicating selection.
- Users must identify which pieces appeared earlier.

"CONTINUE" Button (Bottom Right)

- Once users have made their selections, they press **“CONTINUE”** to confirm their choices and proceed.

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3. How the Selection Phase Works

1. Users analyze the displayed chess pieces.
2. They recall which ones they saw during the timed phase.
3. They select remembered pieces (green dot indicates selection).
4. Clicking “CONTINUE” finalizes their choices and leads to the **Result Page**.

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4. Cognitive & UX Benefits

- Tests memory accuracy by comparing user selections with previously displayed pieces.
- Incorporates **distractors** to evaluate how well users distinguish real vs. false memories.
- Simple interface with clear selection indicators (green dots).
- Gamified elements (color cues, clean layout) make the task engaging and intuitive.

FRAME 5: RESULT PAGE

Analysis of Score & Feedback Screen

- **Purpose:** Displays game outcome (Win/Loss/Draw).
- **Key UI Elements:**
  - Title: “GAME RESULT”
  - Scoreboard showing winner and number of moves.
  - Action Buttons: “RESTART”, “EXIT”, “ANALYZE GAME”.
  - Animated chess mascot (optional).
- **Cognitive & UX Benefits:**
  - Instant feedback helps players track performance.
  - Options give control over next action.
  - Gamified visuals reduce stress and increase engagement.